SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

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www.seewandel.org

Ecomeets @ Eawag, Dübendorf
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Prof. D. Straile



Prof. U. Gaedke



Dr. J. Alexander



Dr. P. Teiber-Sießegger

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

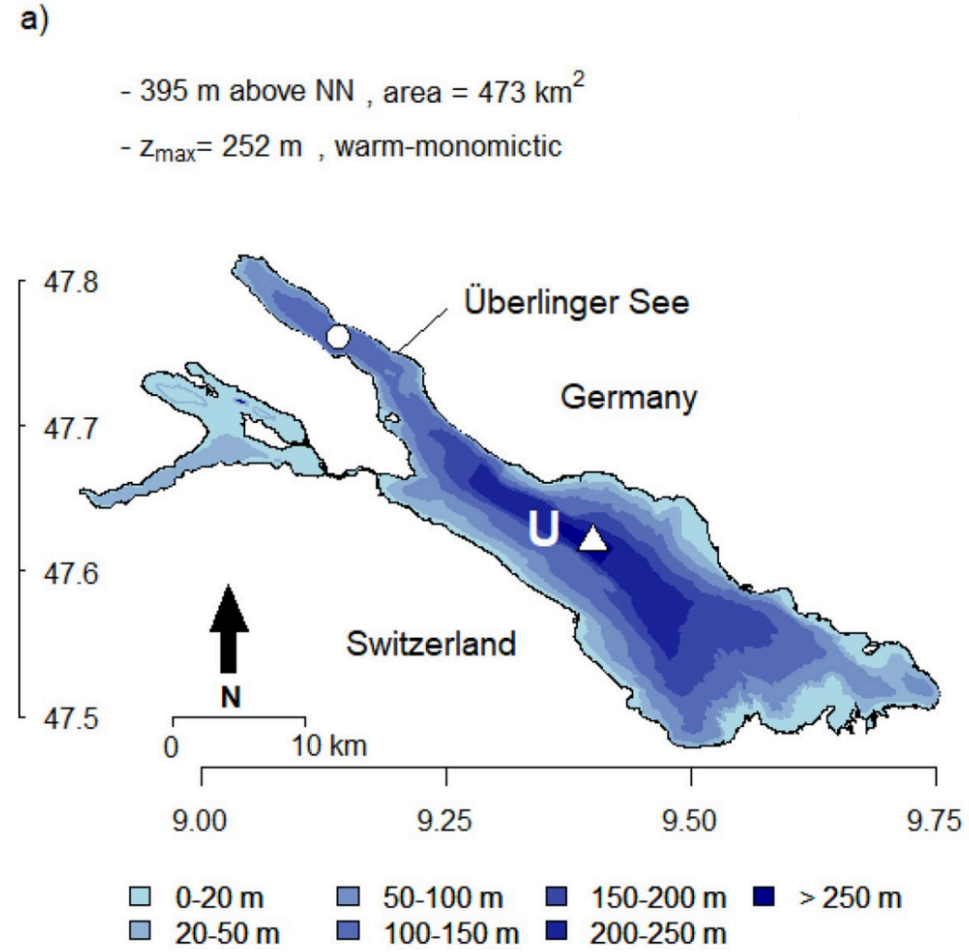
2. Whether they propagate and can be predicted is still open.

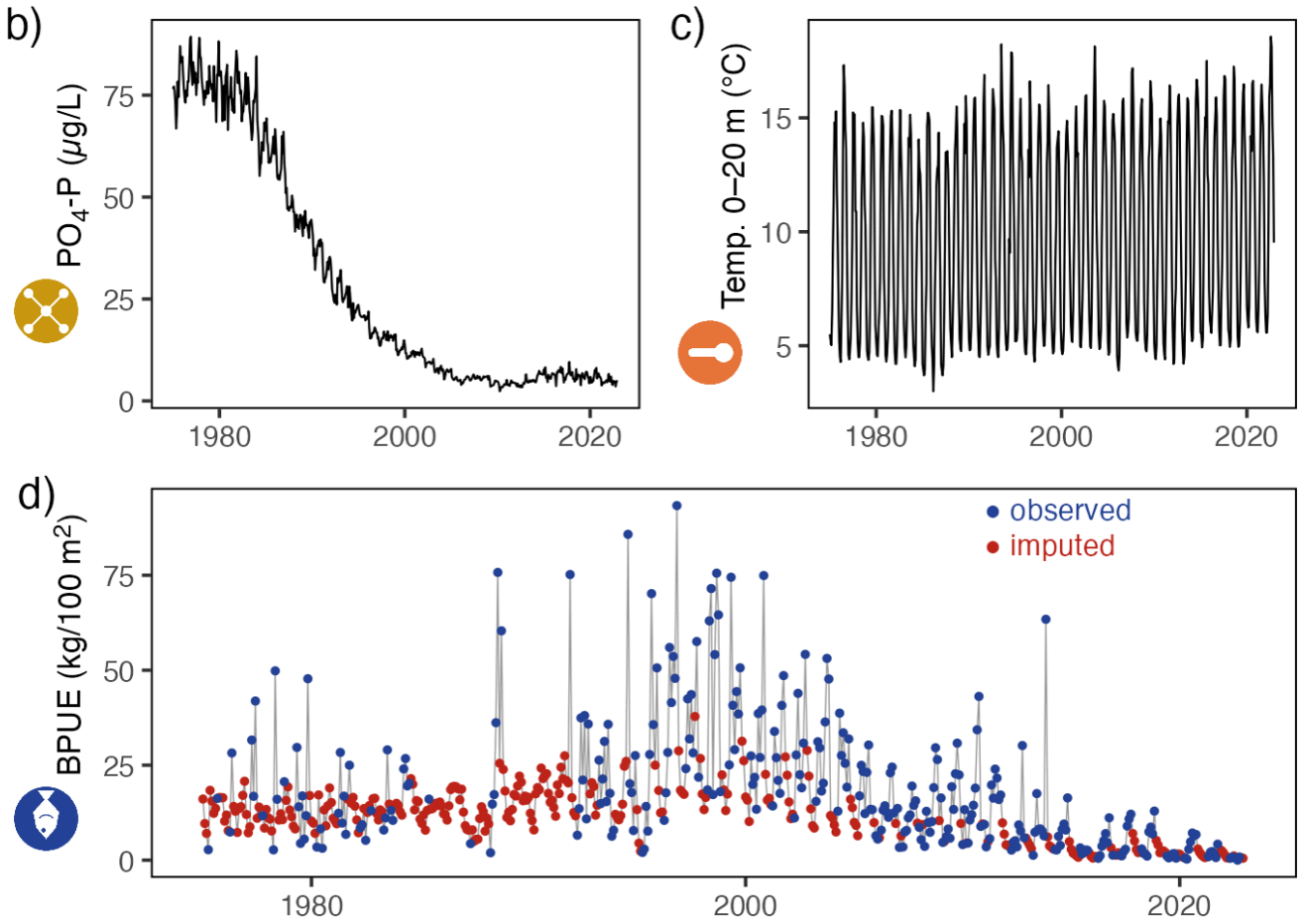


Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background





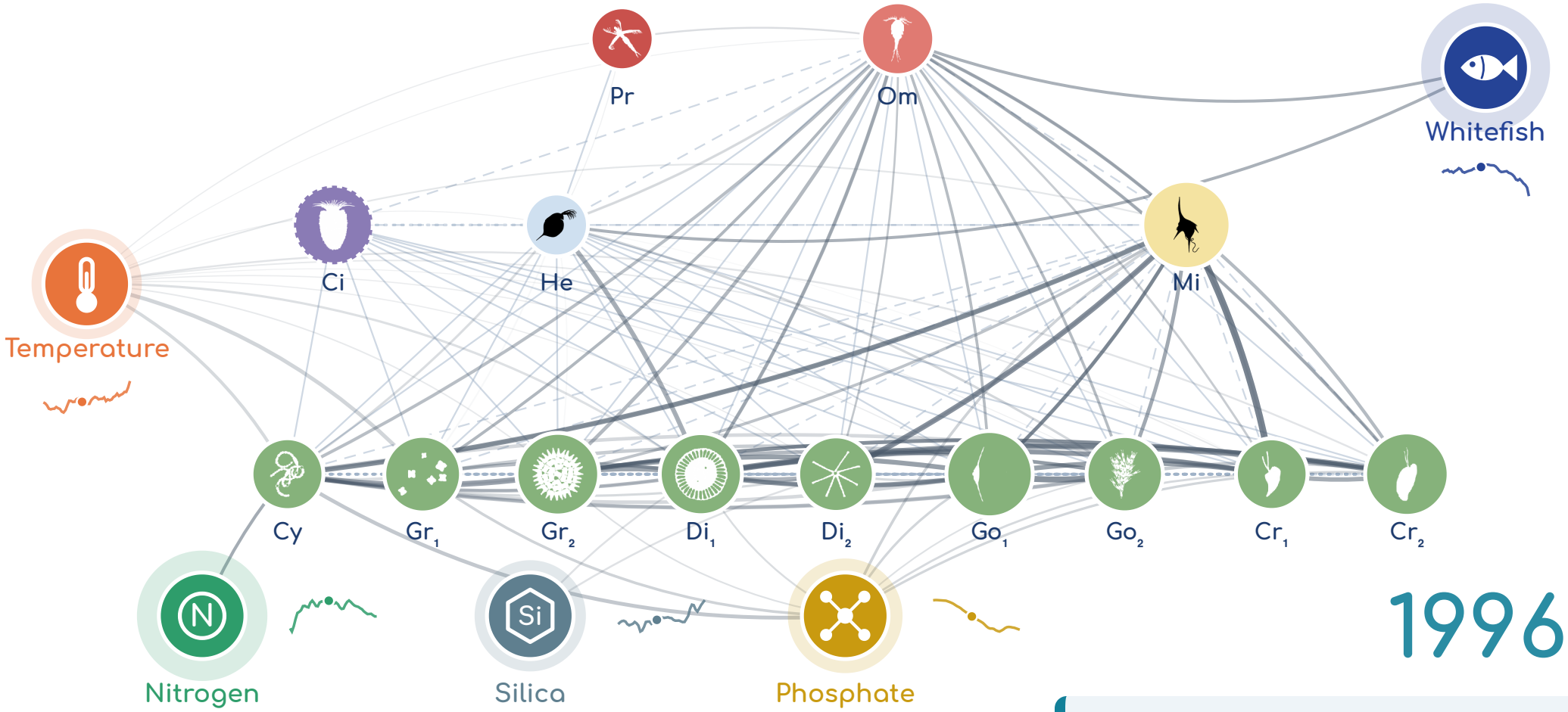
Lake Constance · 473 km²

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

Map (a): Straile 2025 · b–d: monthly IGKB and IBKF data, 1975–2022 (red: imputed)

What we know: the plankton community



1996

Study period: 1975–2022 · monthly time series

The plankton community changes across time

Network modified from Merz et al. 2023, Nat. Clim. Change. Links without a strength: minimum intensity.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

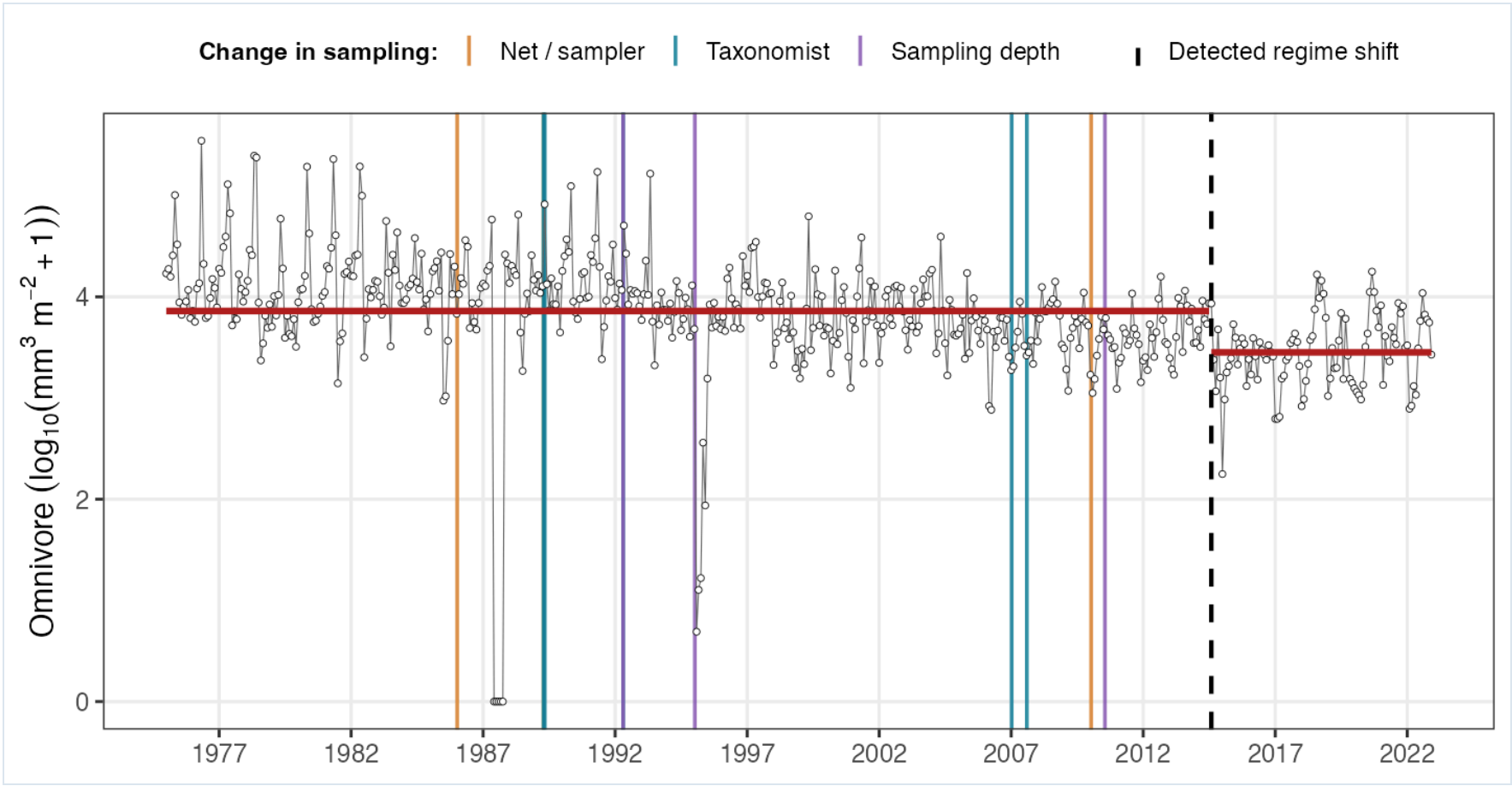
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts





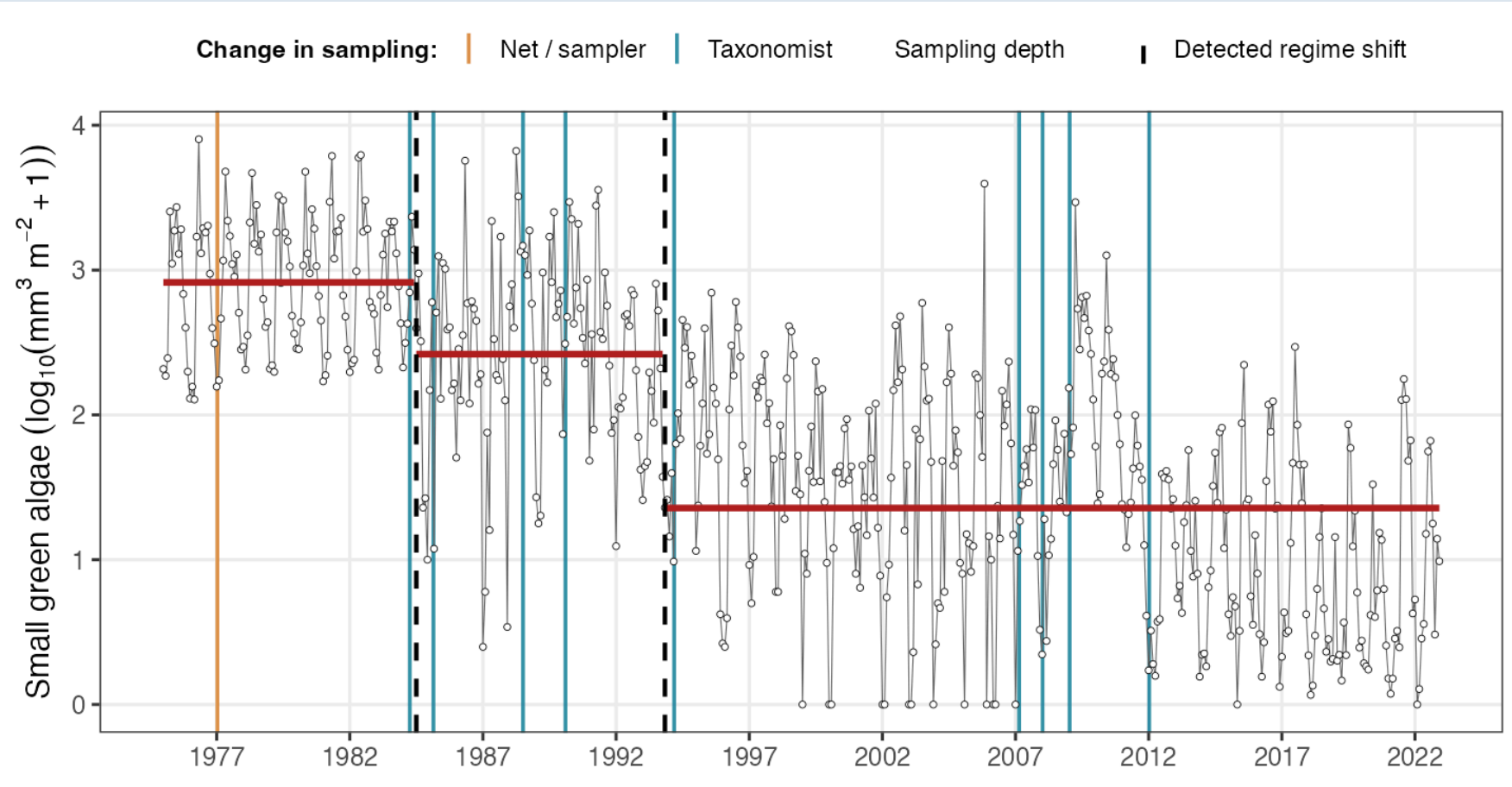
Shift: Aug 2014

Abruptness 1.19
jump size vs. usual swings (> 1 = abrupt)

Omnivores, monthly biovolume. Red: period means. The shift does not coincide with any change in sampling.

The omnivores dropped abruptly in 2014

Results: detection of regime shifts



Small green algae, monthly biovolume. Red: period means.



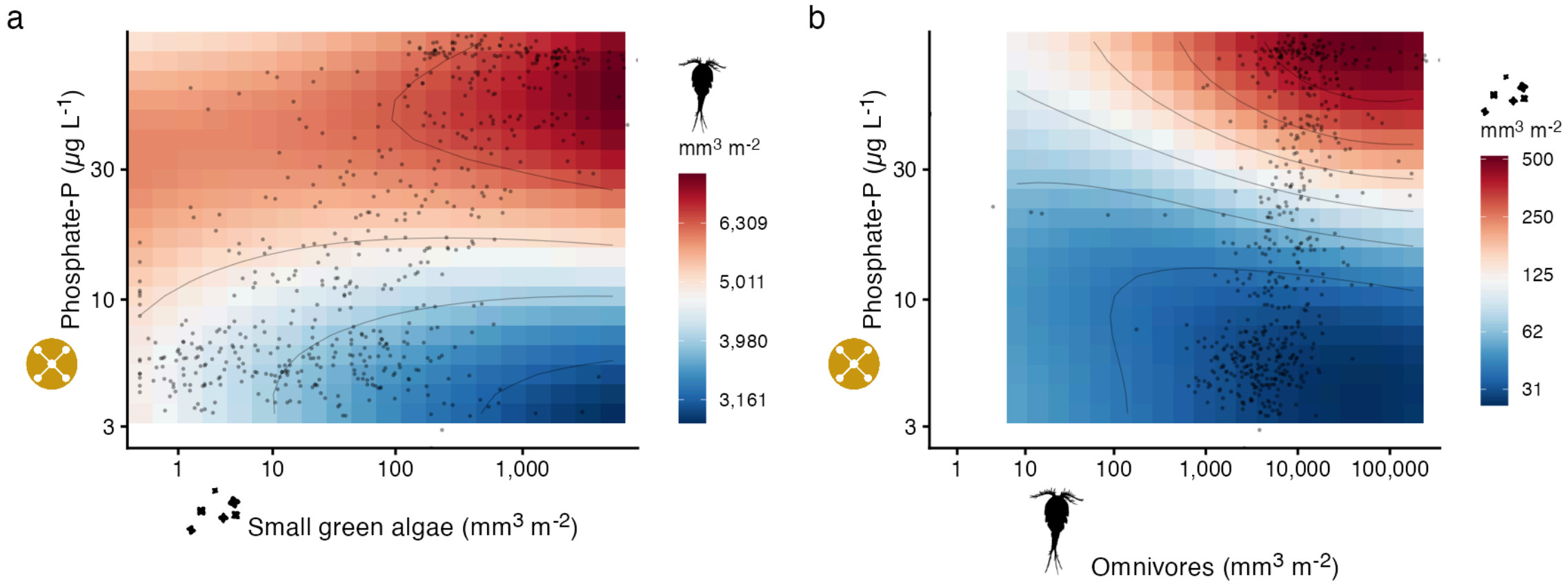
Jul 1984
abruptness 1.20

Nov 1993
abruptness 1.15

Both ≤ 4 months from a taxonomist change

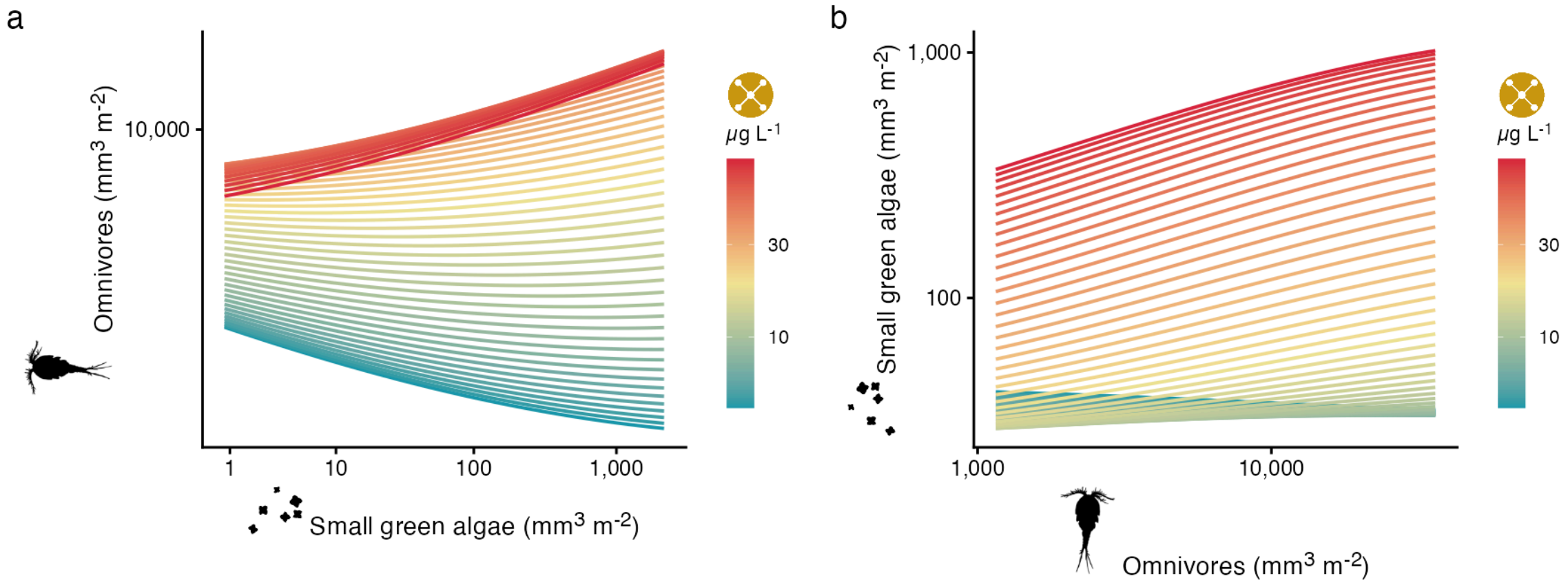
Small green algae shifted twice

Explain regime shifts



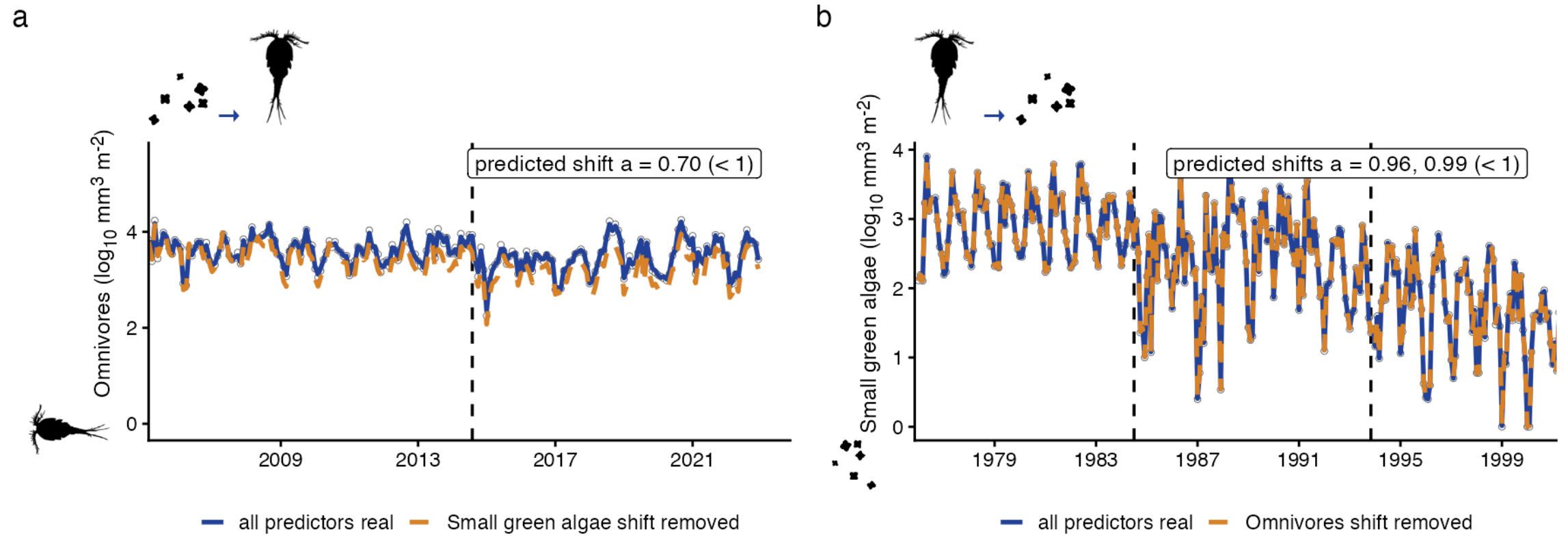
Phosphate drives both guilds

Explain regime shifts: partial dependence



Algae effect depends on phosphate

Results: propagation of regime shift

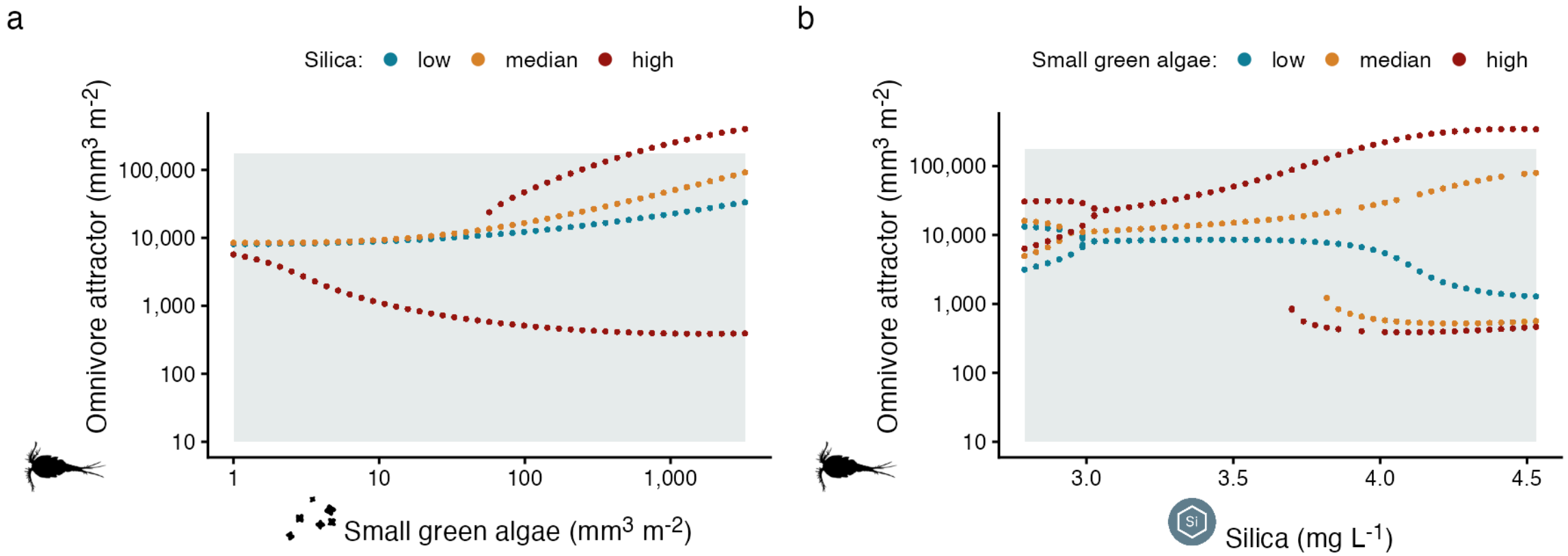


Shift-removal test: **0 of 3** shifts reproduced by the model (a < 1), so none is testable

No evidence of propagation

Regime shift: prediction

🚧 In progress · preliminary result

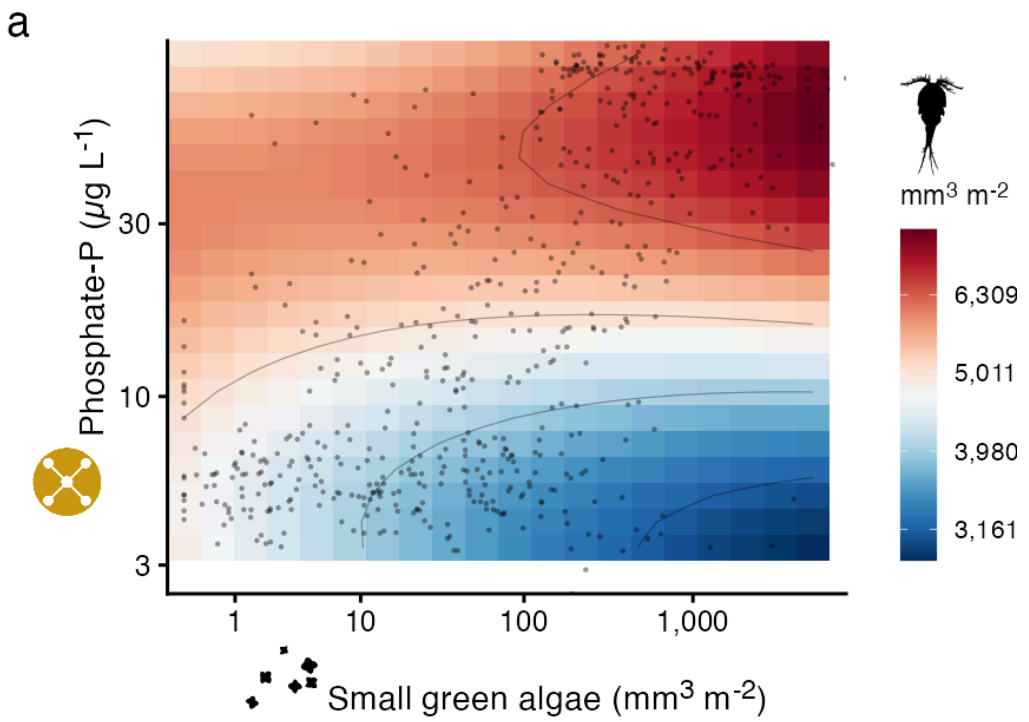


Two stable states at high silica

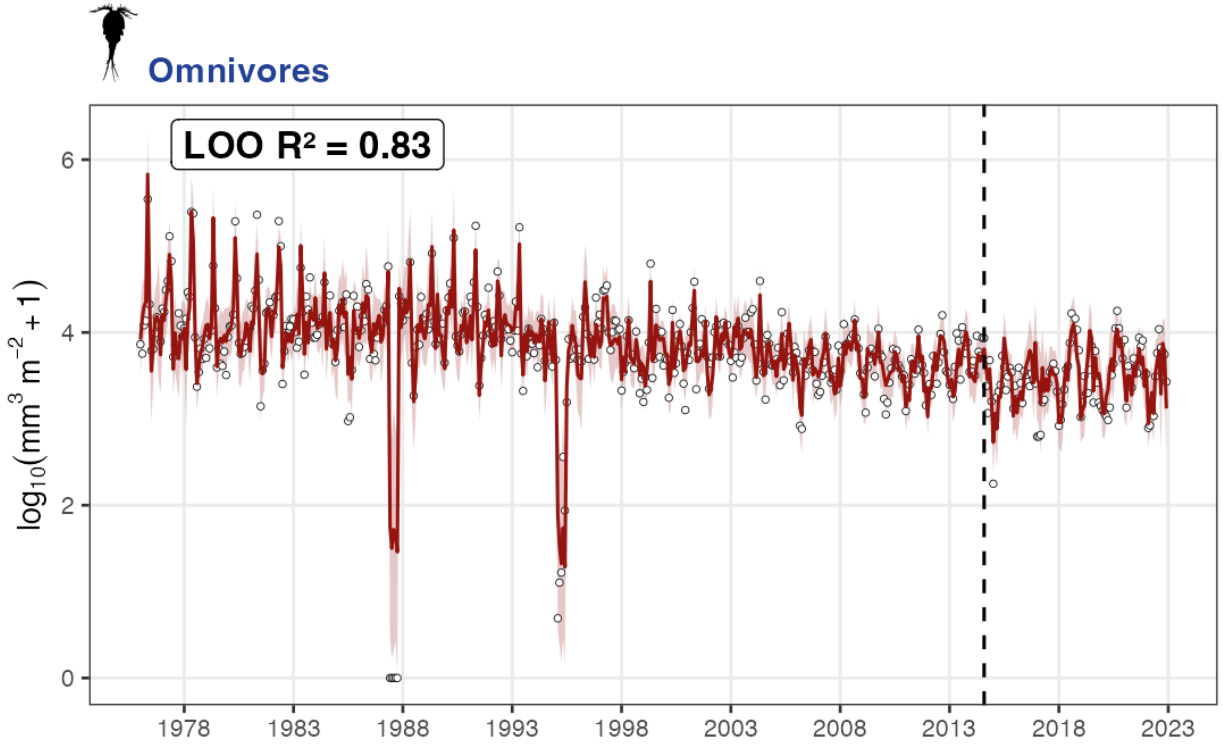
Next steps

1. Refit the model with 2024 and 2025 IGKB data
2. Forecast for the year 2026
3. Manuscript submission: February 2027

Contribution to Interreg outputs



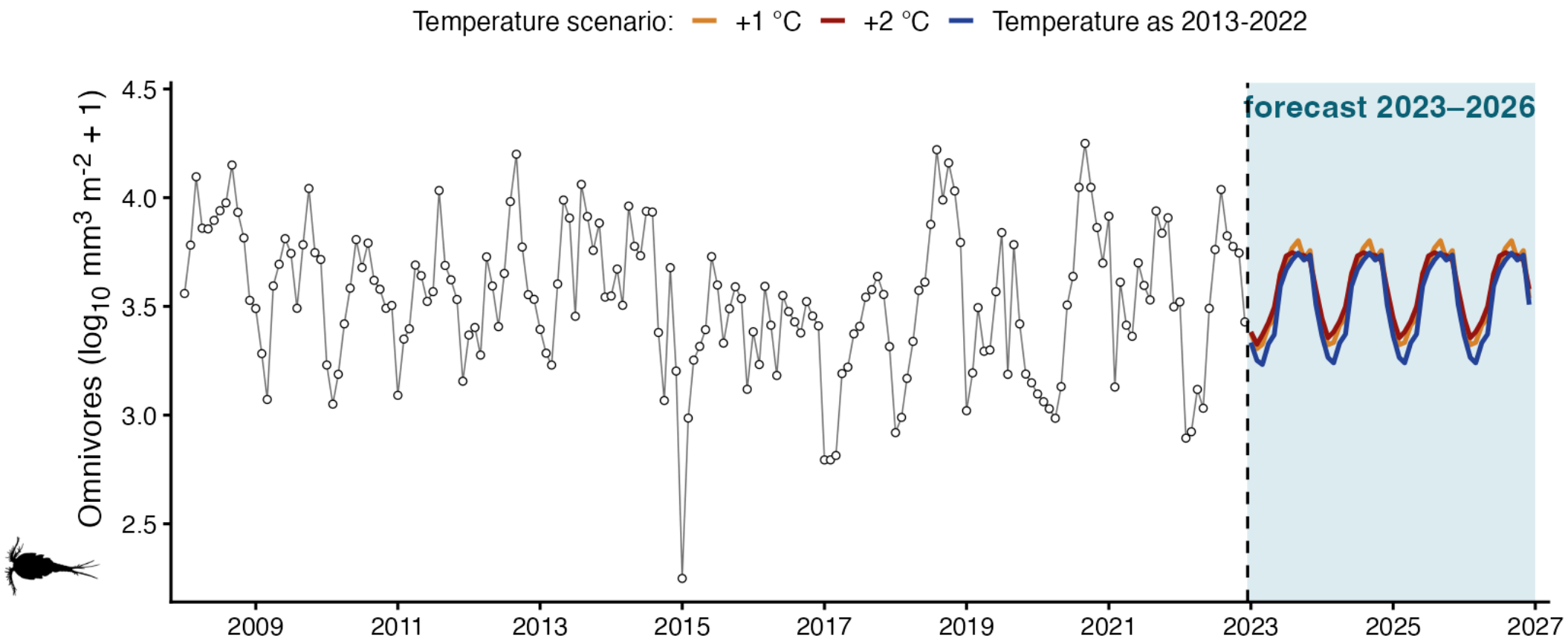
Explain: heat map of the omnivores



Predict: omnivores observed (points) and predicted (line)

Tools to explain and predict plankton

Contribution to Interreg project results



Validation, 12 months ahead (2018–2022): R^2 = **0.31** · ρ = **0.58**

Omnivore forecast under temperature scenarios

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

2. Whether they propagate and can be predicted is still open.



Acknowledgements

SeeWandel Climate

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Kofinanziert von der Europäischen Union

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Prof. D. Straile



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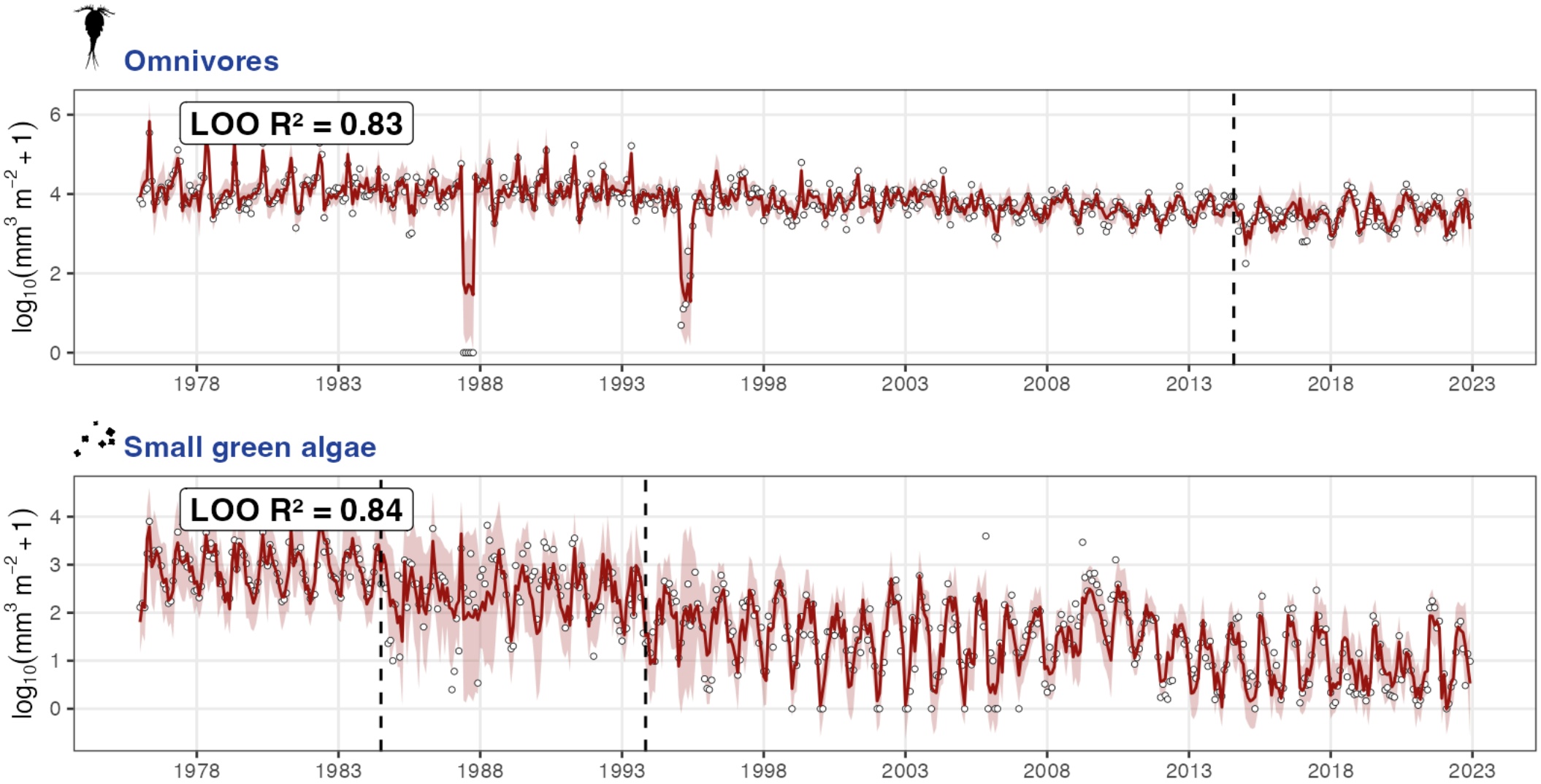


Dr. P. Teiber-Sießegger

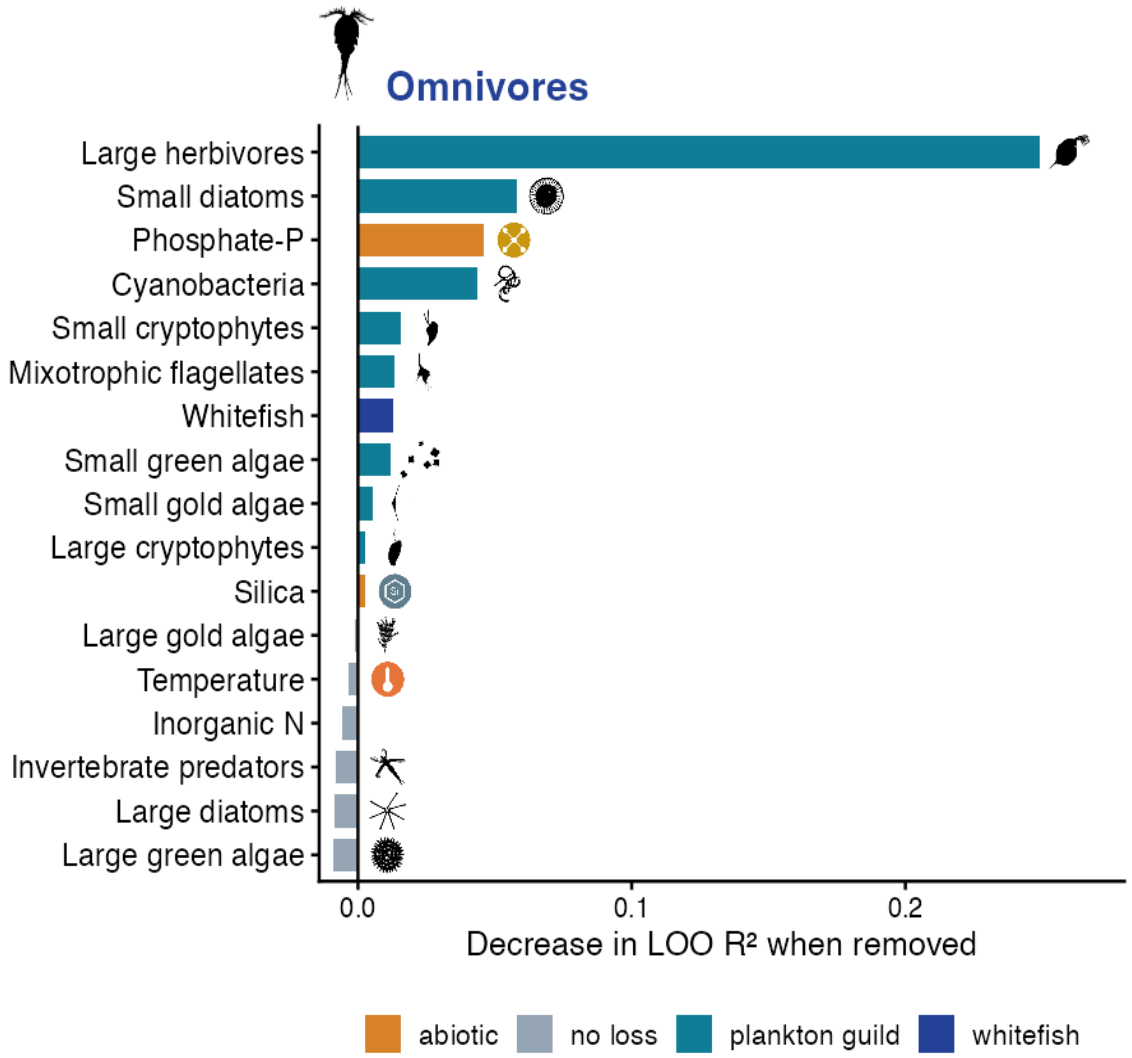
Supplementary materials

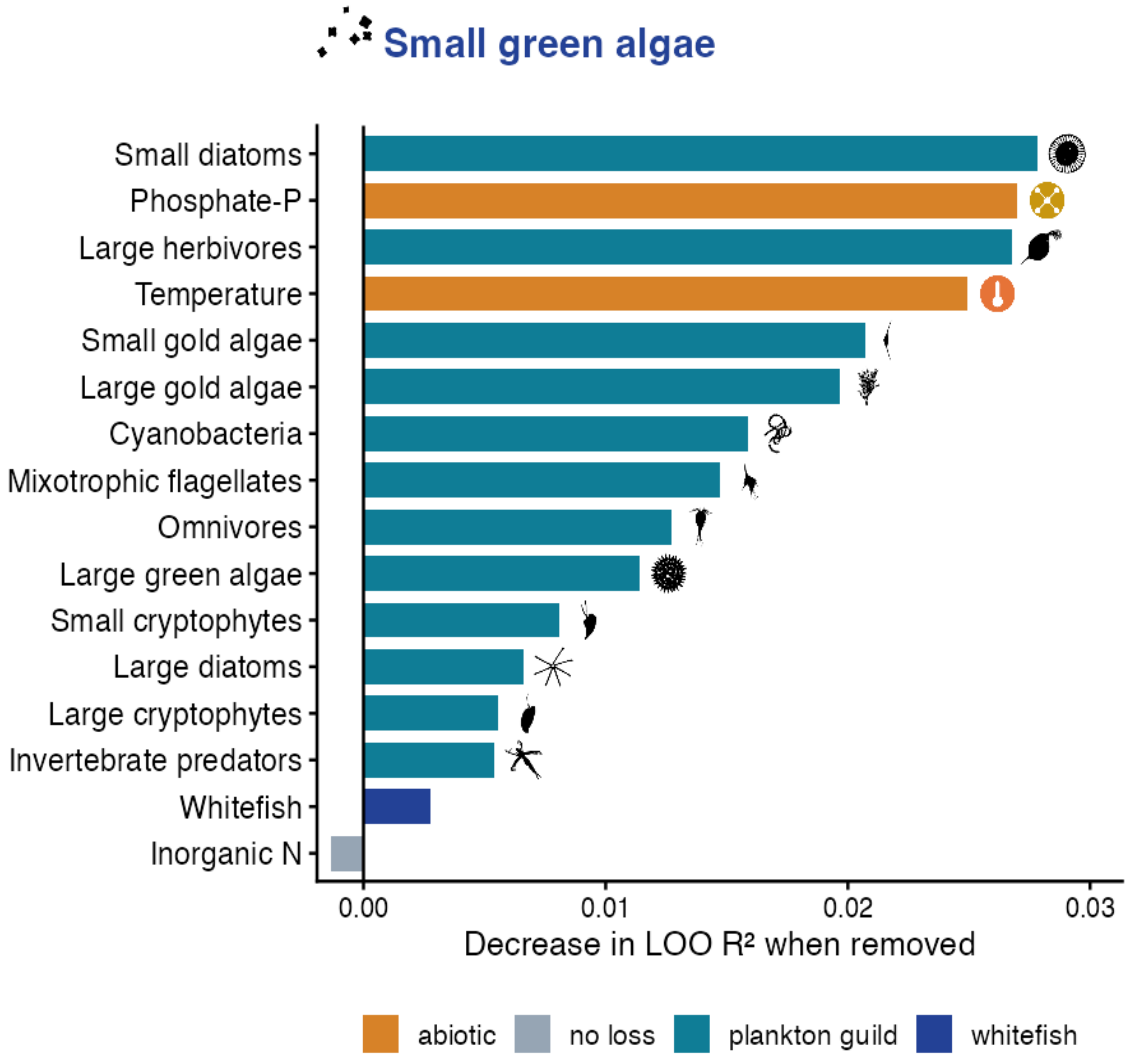


Model fit: Leave-One-Out R^2



Which predictors matter?





Which predictors were used?

Omnivores LOO R^2 = 0.83

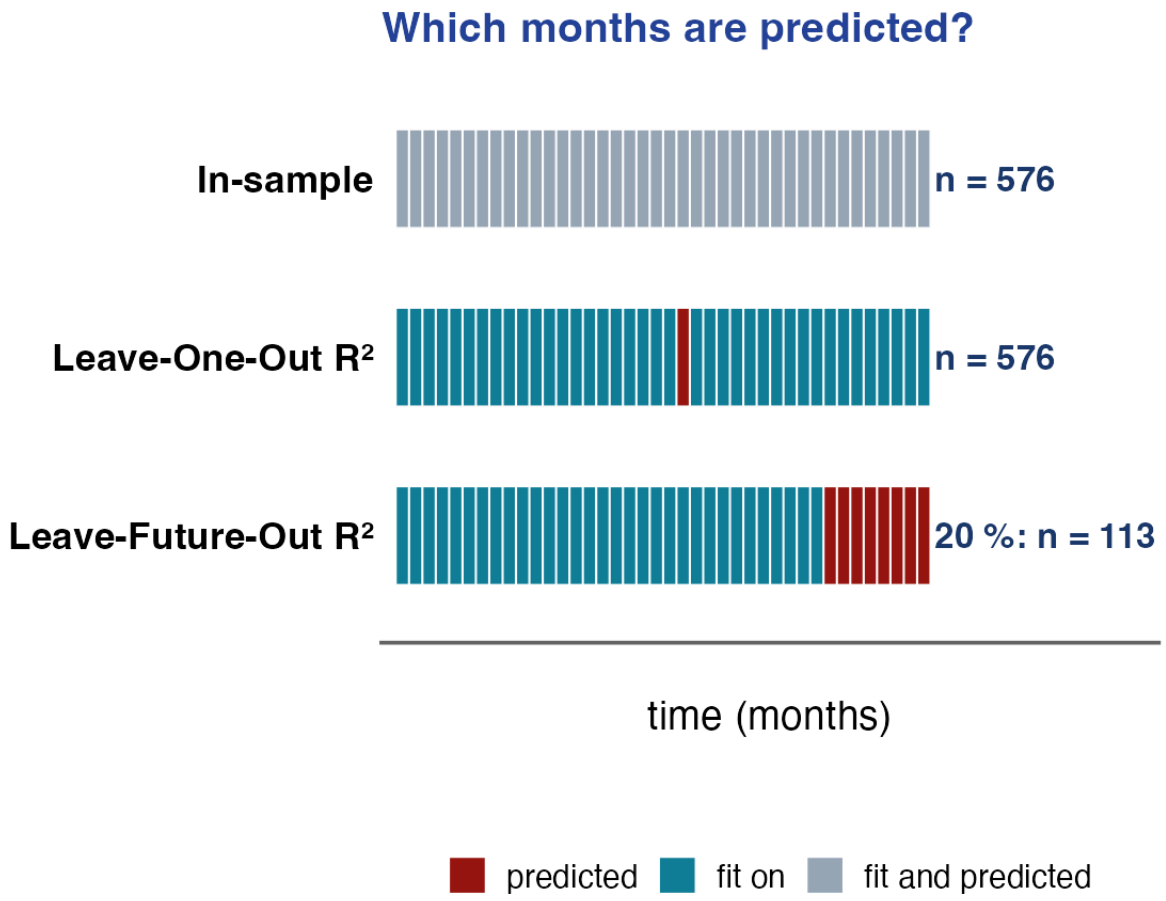
- Phosphate-P — 5 lags
- Large herbivores — 1 lags
- Small diatoms — 2 lags
- Cyanobacteria — 2 lags
- Temperature — 3 lags
- Whitefish — 6 lags
- Invertebrate predators — 3 lags
- Small cryptophytes — 2 lags
- Small green algae — 2 lags
- Large gold algae — 3 lags
- Mixotrophic flagellates — 1 lags
- Small gold algae — 2 lags
- Large green algae — 1 lags
- Large diatoms — 2 lags
- Inorganic N — 1 lags
- Silica — 0 lags
- Large cryptophytes removed by ARD

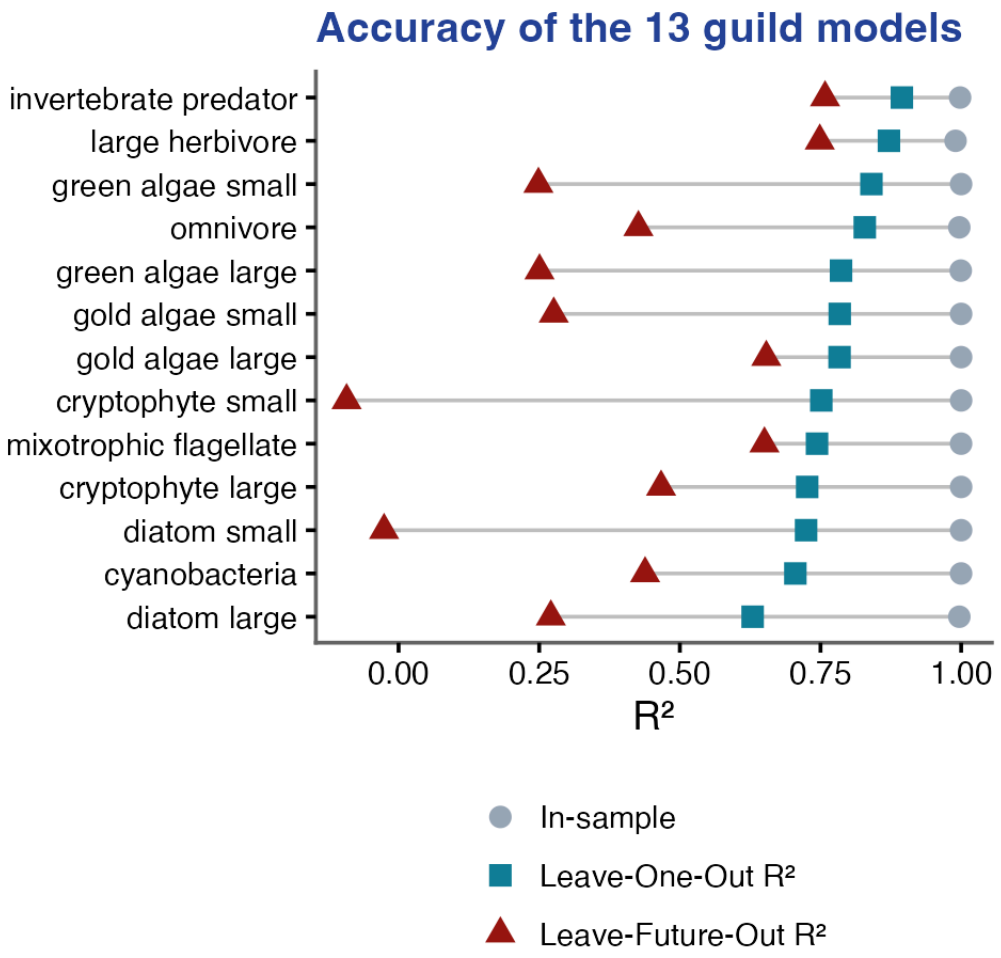
Small green algae LOO R^2 = 0.84

- Phosphate-P — 5 lags
- Temperature — 1 lags
- Large gold algae — 6 lags
- Mixotrophic flagellates — 5 lags
- Large herbivores — 4 lags
- Small diatoms — 3 lags
- Omnivores — 4 lags
- Cyanobacteria — 4 lags
- Large cryptophytes — 4 lags
- Large green algae — 3 lags
- Invertebrate predators — 3 lags
- Small gold algae — 2 lags
- Small cryptophytes — 1 lags
- Whitefish — 3 lags
- Large diatoms — 1 lags
- Inorganic N removed by ARD
- Silica not entered

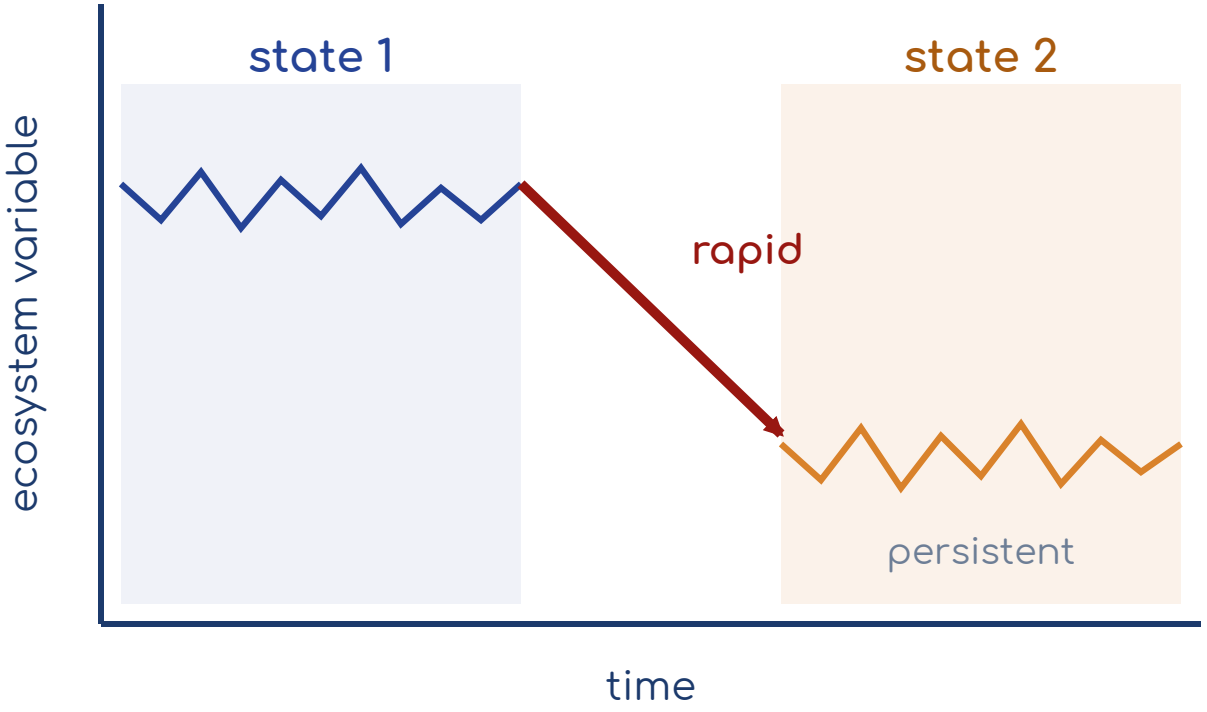
Candidate predictors: all other guilds, temperature, phosphate, silica, inorganic N and whitefish, each at lags 0–12 months

In-sample versus out-of-sample R²





What is a regime shift?



- Variable **jumps** rapidly
- Faster than **background** change
- **New state** persists
- Found by **change points**
- **Drivers** not considered

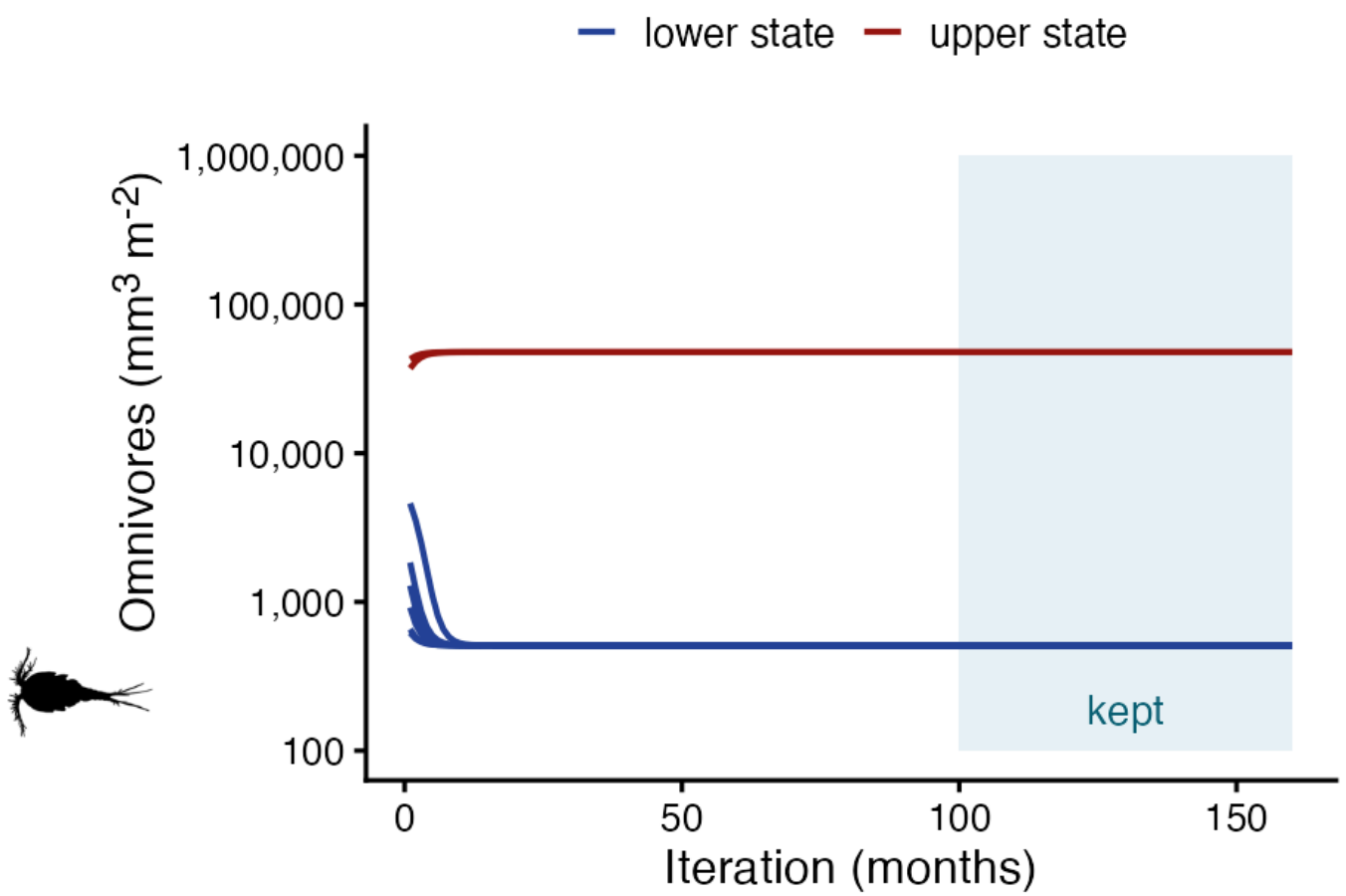
Ratajczak et al. 2018 · Folke et al. 2004

How the bifurcation diagram is made

1. **Fit GP-EDM** — omnivore model
2. **Fix drivers** — temperature, phosphate
3. **Pick driver level** — green algae, silica
4. **Random start** — two past months
5. **Predict** — feed it back
6. **Repeat 160 times** — keep last 60
7. **Plot** — kept values vs driver

Steps 4–6 for one setting: **next slide ➜**

Steps 4–6: iterating the model



4 Random starts

5 Predict, feed back

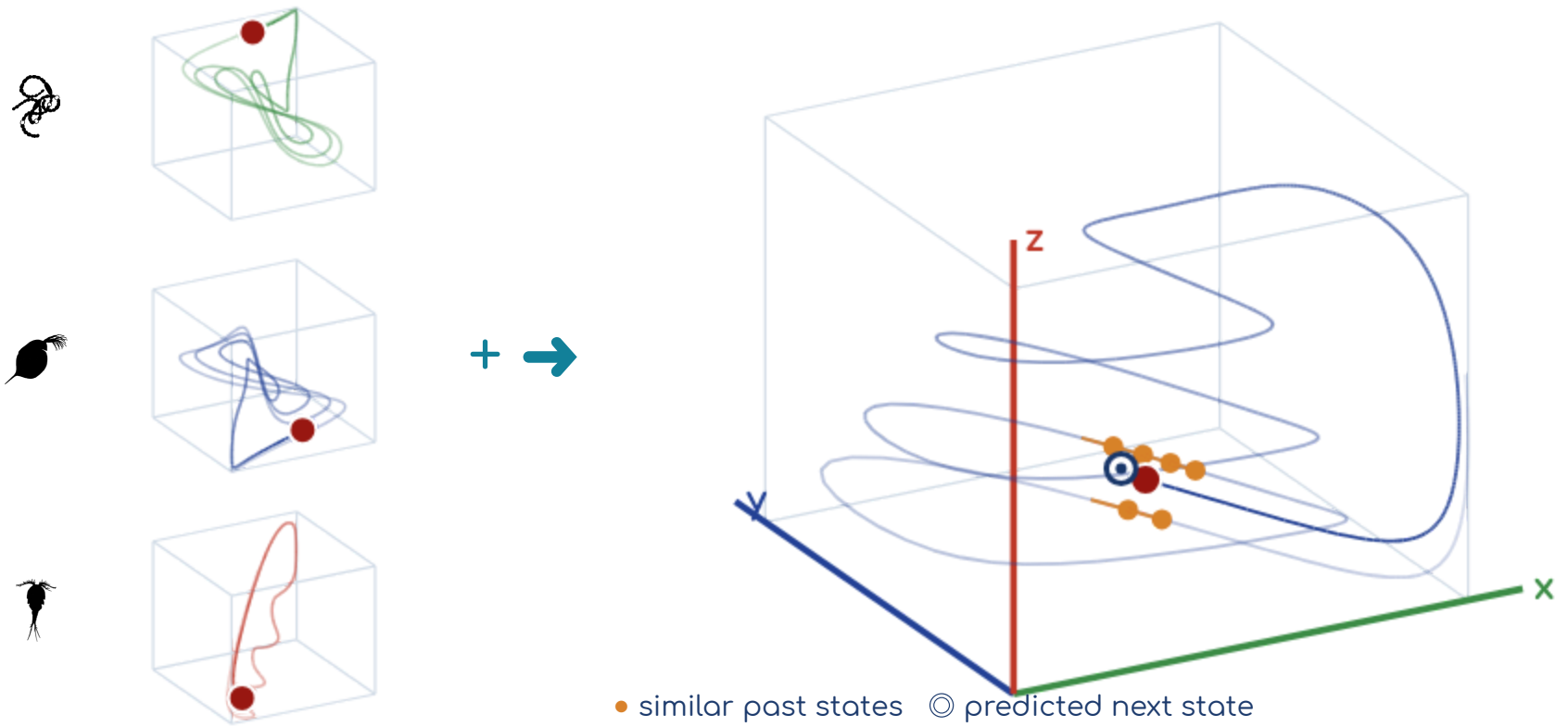
6 Keep last 60 months

Starts reach **two** states: bistability

7 Each end state = one point in the diagram, at this driver level

Example: small green algae 100 mm³ m⁻², high silica

GP-EDM: multivariate embedding



All species + lags

One GP model

ARD weights inputs

Dynamics

$x_{t+1} = f(x_t, y_t, z_t, x_{t-1}, y_{t-1}, z_{t-1}, \dots) + \varepsilon_t$

GP kernel

$k = \sigma^2 \exp(-\sum_i \phi_i (u_i - u'_i)^2)$

Simulated three-species food chain; design after Ye & Sugihara 2016

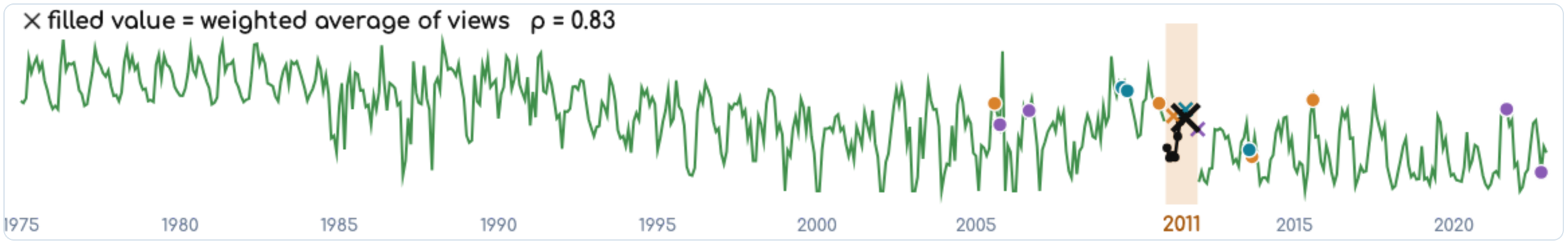
Multiview EDM: the core idea

Target: small green algae, not sampled in 2011 — zooplankton and abiotic variables give the views

View 1 t, t−2 t−1 — ρ = 0.82

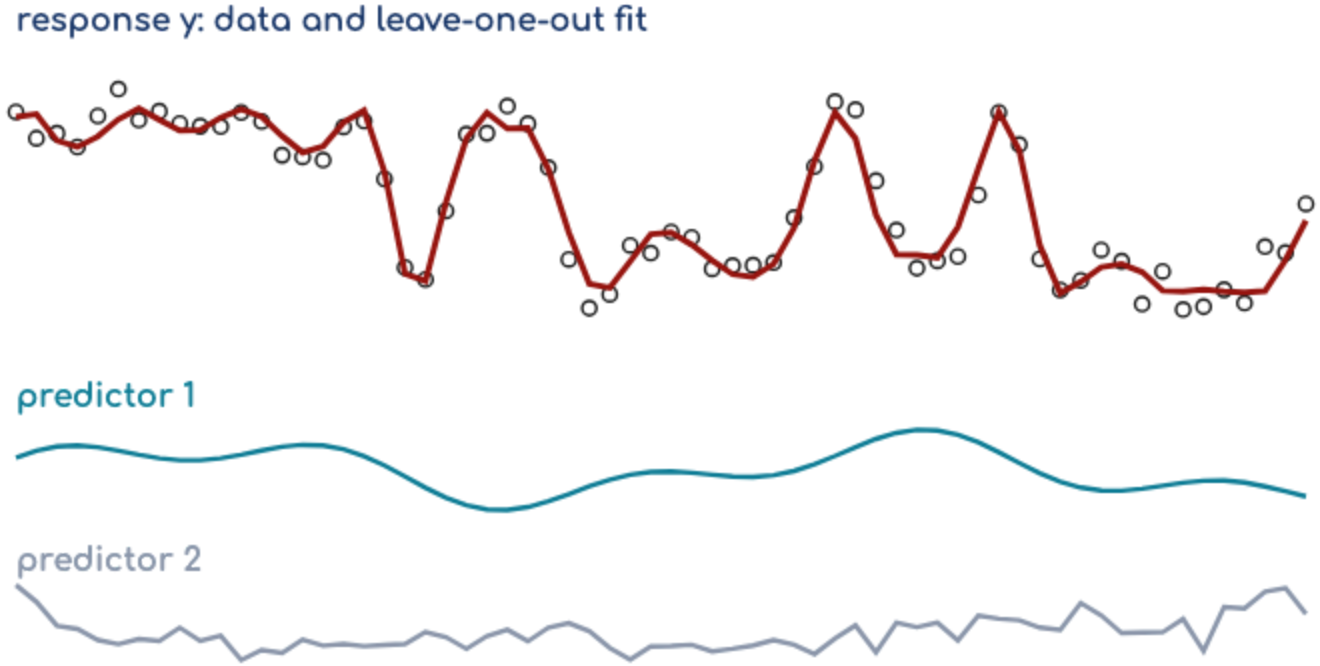
View 2 t, t−1 t−3 — ρ = 0.79

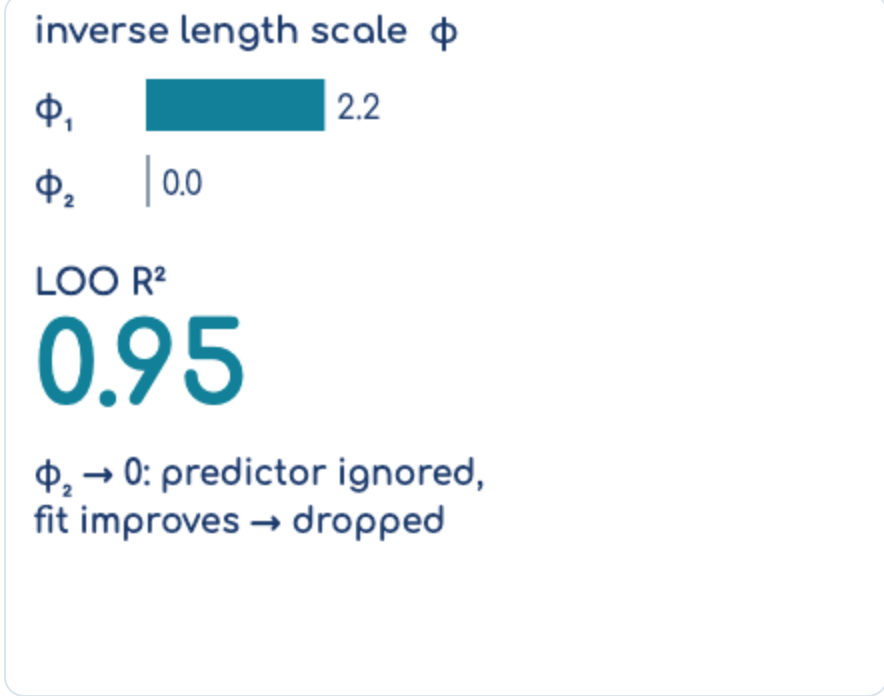
View 3 t, t−3 t−2 — ρ = 0.76



Real data, Lake Constance (IGKB). Grey: past states · red: month in 2011 · coloured: similar states · design after Carpenter et al. 2025 (Fig. 1)

Automatic relevance determination (ARD)





One ϕ per predictor

$k(x,x') = \sigma^2 \exp(- \phi_1(x_1 - x'_1)^2 - \phi_2(x_2 - x'_2)^2)$

Why drop predictors?

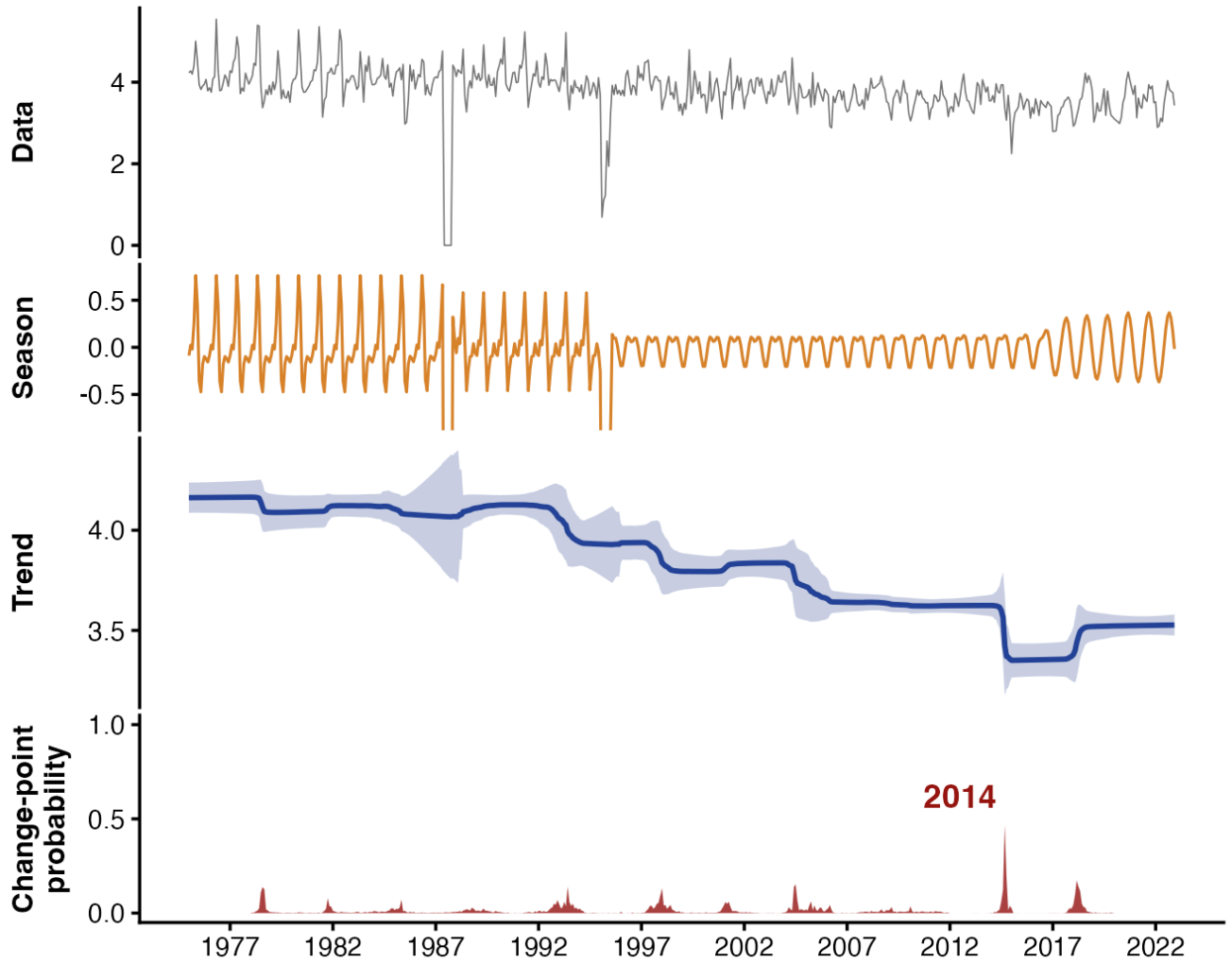
$\phi_2 = 0 \Rightarrow x_2$ never changes k

$\Rightarrow$ less noise, better LOO fit

Rule

$\phi_i \leq 0.001$: dropped

What does BEAST do?



Splits the series

y_t = trend + season + noise

Finds change points

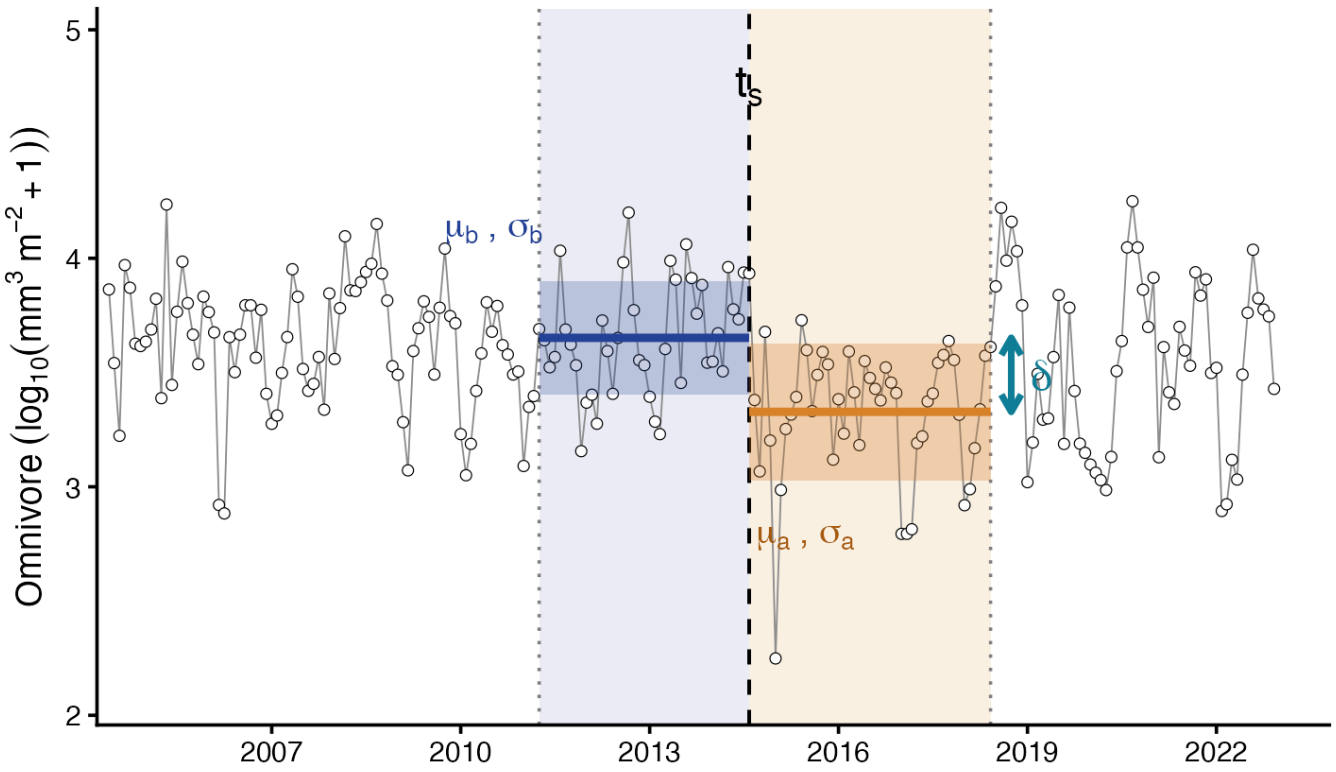
jumps of the **trend**

Gives a probability

Pr(change point) per month

Illustration: BEAST on the omnivores alone

Detecting regime shifts: criteria



1 Difference of the two chunks

$$\delta = \mu_a - \mu_b \quad 0 \notin CI_{95\%}(\delta)$$

CI = [0.19, 0.42] ✓

2 Abruptness: jump vs. usual swings

$$a = \frac{|\mu_a - \mu_b|}{½(\sigma_b + \sigma_a)} > 1$$

a = 1.19 ✓

Kept if 1 and 2 hold

Which genera? Zooplankton and mixotrophs



Invertebrate predators

Pr

Leptodora	73%
Bythotrephes	27%



Omnivores

Om

Cyclops	80%
Mesocyclops	17%
Acanthocyclops	1%
Thermocyclops	1%
Asplanchna	0%

+ 4 more



Large herbivores

He

Daphnia	91%
Eudiaptomus	6%
Bosmina	2%
Diaphanosoma	1%
Acanthodiaptomus	0%

+ 6 more



Mixotrophic flagellates

Mi

Ceratium	46%
Gymnodinium	45%
Peridinium	7%
other Flagellaten	1%
other Phytoflagellaten	0%

+ 14 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (1/2)



Cyanobacteria

Cy colonial

Anabaena	44%
Aphanizomenon	16%
Planktothrix	9%
Cyanothece	7%
Aphanothece	6%

+ 25 more



Green algae, small

Gr1 ≤ 550 µm³

Volvocales	30%
unidentified algae	24%
Chlorella	21%
Oocystis	7%
Characium	3%

+ 21 more

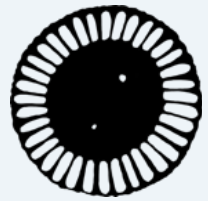


Green algae, large

Gr2 > 550 µm³ or colonial

other Chlorococcales	23%
Pandorina	18%
Mougeotia	16%
Chlamydomonas	10%
Ulothrix	6%

+ 46 more

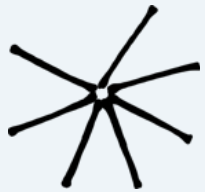


Diatoms, small

Di1 ≤ 800 µm³

Stephanodiscus	81%
Cyclotella	17%
Achnanthes	1%
Fragilaria	1%
Nitzschia	0%

+ 9 more



Diatoms, large

Di2 > 800 µm³ or colonial

Fragilaria	50%
Stephanodiscus	15%
Asterionella	14%
Cyclotella	10%
Tabellaria	4%

+ 18 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (2/2)



Gold algae, small

Go1 ≤ 250 µm³

Erkenia	63%
Chrysochromulina	23%
Chromulina	4%
Ochromonas	2%
Pseudokephyrion	2%

+ 13 more



Gold algae, large

Go2 > 250 µm³ or colonial

Dinobryon	42%
Mallomonas	17%
Uroglena	12%
Chrysophyceae	12%
Pseudopedinella	6%

+ 9 more



Cryptophytes, small

Cr1 ≤ 700 µm³

Rhodomonas	99%
Plagioselmis	1%
Cyathomonas	0%
Cryptaulax	0%
Chilomonas	0%

+ 1 more



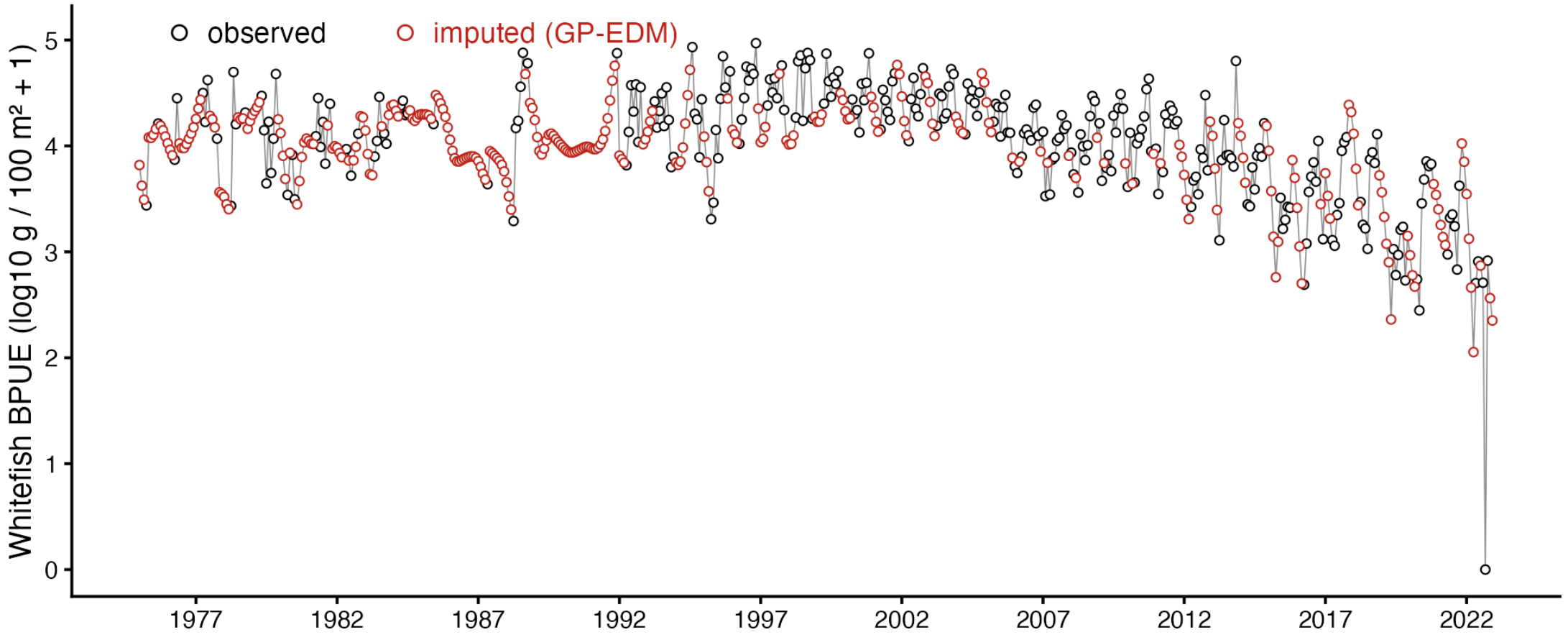
Cryptophytes, large

Cr2 > 700 µm³

Cryptomonas	100%
other Cryptophyceen	0%

Top genera by share of biovolume, IGKB 1975–2022

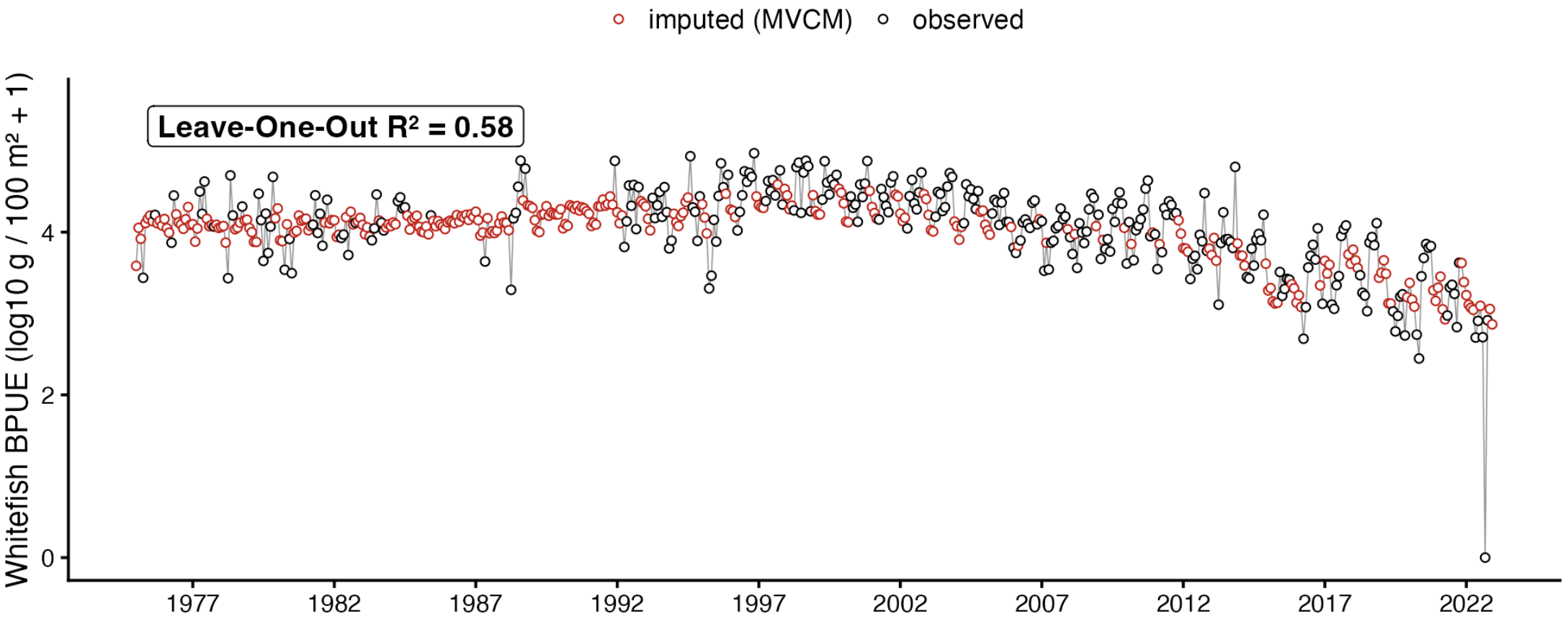
Whitefish time series: observed and imputed



287 of 576 months imputed (50 %)

GP-EDM, LOO R^2 = **0.55**

Whitefish time series: imputed with MVCM



287 of 576 months imputed

MVCM: Carpenter et al. 2025

GP-EDM version: Leave-One-Out R² = **0.55**

MVCM: multiview cross-mapping

1 Make views

4 inputs each, lags −2 to +2

2 Cross-map

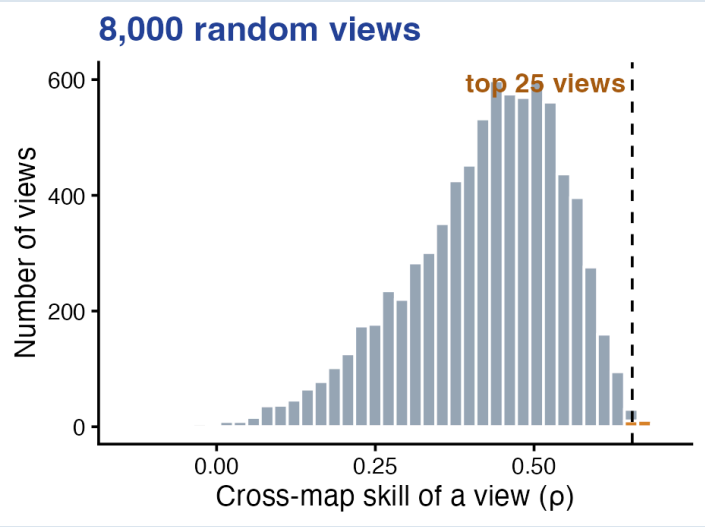
5 similar months, weighted

3 Rank views

by skill ρ

4 Average top 25

weights ∝ ρ



$\hat{y}_t = \sum_v w_v \hat{y}_{v,t} \quad w_v = \rho_v / \sum \rho$

8,000 random views

Simplex variant of MVCM

Carpenter et al. 2025; Ye & Sugihara 2016

References (1/2)

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Map in slide 6 (panel a): Straile 2025. Panels b–d: monthly IGKB (phosphate, temperature) and IBKF (whitefish) data, 1975–2022.